SECTION 1 Time—30 minutes 40 Questions

For each question in this section, choose the best answer and fill in the corresponding oval on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. SURPLUS: (A) shortage (B) criticism (C) heated argument (D) sudden victory (E) thorough review

2. VAGUE: (A) abundant (B) correct (C) definite (D) stable (E) petty

3. COLLABORATE: (A) pay attention (B) issue orders (C) become simpler (D) gain strength (E) work alone

4. NONCOMBATANT: (A) forerunner (B) inhabitant (C) eccentric (D) warrior (E) expert

5. POROUS: (A) inclement (B) impermeable (C) concentrated (D) perishable (E) uncongealed

6. EXTRADITE: (A) stimulate (B) underemphasize (C) become specialized (D) withhold wages from (E) grant asylum to

7. SOPHOMORIC: (A) charitable (B) profound (C) accidental (D) histrionic (E) straightforward

8. DEBILITATE: (A) reveal (B) decorate (C) sustain (D) put forth (E) give approval

9. PRODIGAL: (A) antagonist (B) successor (C) grouch (D) underweight person (E) thrifty individual

10. PRECOCIOUS: (A) elusive (B) isolated (C) understandable (D) proven (E) backward

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that, when inserted in the sentence, best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

11. The summer's ---- drought has brought about a ---- shedding of leaves.

(A) torrid. .nonexistent
(B) inconsequential. .disastrous
(C) rainy. .colorful
(D) severe. .premature
(E) unusual. .traditional

12. At the age of forty-five, with a worldwide reputation and an as yet unbroken string of notable successes to her credit, Carson was at the ---- of her career.

(A) paradigm (B) zenith (C) fiasco (D) periphery (E) inception

13. The fact that they cherished religious objects more than most of their other possessions ---- the ---- role of religion in their lives.

(A) demonstrates. .crucial
(B) obliterates. .vital
(C) limits. .daily
(D) concerns. .informal
(E) denotes. .varying

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14. The alarm voiced by the committee investigating the incident had a ---- effect, for its dire predictions motivated people to take precautions that ---- an ecological disaster.

(A) trivial. .prompted
(B) salutary. .averted
(C) conciliatory. .supported
(D) beneficial. .exacerbated
(E) perverse. .vanquished

15. After the First World War, a large decrease in Mexican immigration to the United States occurred because the wartime ---- of immigration restrictions was ----.

(A) waiver. .rescinded
(B) enactment. .overturned
(C) abolition. .endorsed
(D) repeal. .guaranteed
(E) execution. .relaxed

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

16. FOOTSTOOL : FOOT :: (A) desk : floor
(B) table : hand (C) bench : chair
(D) pillow : head (E) pencil : pocket

17. CANOE : RAPIDS :: (A) plane : turbulence
(B) truck : garage (C) oar : rowboat
(D) factory : automation (E) pond : stream

18. IRON : BLACKSMITH :: (A) gold : miser
(B) clay : potter (C) food : gourmet
(D) steel : industrialist (E) silver : miner

19. ACRE : LAND :: (A) distance : space
(B) speed : movement (C) gallon : liquid
(D) degree : thermometer (E) year : birthday

20. CONVENE : ASSEMBLY :: (A) borrow : library
(B) reprove : defiance (C) contrast : shadow
(D) implicate : court (E) compile : collection

21. COMPATRIOTS : COUNTRY ::
(A) transients : home
(B) kinsfolk : family
(C) competitors : team
(D) performers : audience
(E) figureheads : government

22. ENTREATY : REQUEST ::
(A) conviction : opinion
(B) theory : truth
(C) success : persistence
(D) debate : compromise
(E) hope : future

23. COVEN : WITCHES :: (A) tavern : bartenders
(B) altar : clergy (C) amulet : vampires
(D) castle : royalty (E) choir : singers

24. STALWART : STRENGTH ::
(A) penniless : wealth (B) portly : girth
(C) persuasive : depth (D) pernicious : health
(E) precious : mirth

25. OPALESCENT : IRIDESCENCE ::
(A) magnetic : repulsion
(B) garish : drabness
(C) flushed : ruddiness
(D) effervescent : stagnation
(E) fluorescent : darkness

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Each passage below is followed by questions based on its content. Answer the questions following each passage on the basis of what is stated or implied in that passage.

"What's that cry of deadly pain?" his sister drawled from the back seat. "Oh, it's you," she said. "Well, well, we have the artist with us again. How utterly utterly." She had a decidedly nasal voice.

He didn't answer her or turn his head. He had learned that much. Never answer her.

"Mary George!" his mother said sharply. "Asbury is sick. Leave him alone."

"What's wrong with him?" Mary George asked.

"There's the house!" his mother said as if they were all blind but her. It rose on the crest of the hill—a white two-story farmhouse with a wide porch and pleasant columns. She always approached it with a feeling of pride and she had said more than once to Asbury, "You have a home here that half those people up there would give their eyeteeth for!"

She had been once to the terrible place where he lived in New York. They had gone up five flights of dark stone steps, past open garbage cans on every landing, to arrive finally at two damp rooms and a closet with a toilet in it. "You wouldn't live like this at home," she had muttered.

"No!" he'd said with an ecstatic look, "it wouldn't be possible!"

She pulled the car into the side drive. "Home again, home again jiggity jig!" she said.

Asbury groaned.

"The artist arrives at death row," Mary George said.

26. Which of the following best describes the tone of the question in line 1 ?

(A) Despair (B) Confusion (C) Ridicule (D) Genuine alarm (E) Cautious curiosity

27. In context, which of the following best describes the feelings expressed by Asbury's statement in lines 22-23 ?

(A) Deranged hatred
(B) Defiant elation
(C) Resentful anger
(D) Defeated resignation
(E) Apathetic satisfaction

28. Based on the information in the passage, which of the following best describes the mother?

(A) Angry at her children's cynical and bickering manner but envious of their intelligence and sharp wit
(B) Aware of the individuality of each of her children and approving of their life-styles
(C) Anxious about her children's interests and welfare and convinced that she knows what is best for them
(D) Somber in viewing her life and pessimistically dwelling on the unpleasant and the depressing
(E) Successful in achieving her goals in life but unconcerned with what others think about her family

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The reading passages in this test are brief excerpts or adaptations of excerpts from published material. The ideas contained in them do not necessarily represent the opinions of the College Board or Educational Testing Service. To make the text suitable for testing purposes, we may in some cases have altered the style, contents, or point of view of the original.

1

From a vantage point in space, an observer could see that the Earth is engaged in a variety of motions. First, there is its rotation on its own axis, causing the alternation of day and night. This rotation, however, is not altogether steady. Primarily because of the Moon's gravitational action, the Earth's axis wobbles like that of an ill-spun top. In this motion, called "precession," the North and South Poles each trace out the base of a cone in space, completing a circle every 25,800 years. In addition, as the Sun and the Moon change their positions with respect to the Earth, their changing gravitational effects result in a slight "nodding" of the Earth's axis, called "nutation," which is superimposed on precession. The Earth completes one of these "nods" every 18.6 years.

The Earth also, of course, revolves around the Sun, in a 6-million-mile journey that takes 365.25 days. The shape of this orbit is an ellipse, but it is not the center of the Earth that follows the elliptical path. Earth and Moon behave like an asymmetrical dumbbell, and it is the center of mass of this dumbbell that traces the ellipse around the Sun. The center of the Earth-Moon mass lies about 3,000 miles away from the center of the Earth, and the Earth thus moves in an S-curve that crosses and recrosses its orbital path. Then too, the Earth accompanies the Sun in the Sun's movements: first, through its local star cloud, and second, in a great sweep around the "hub" of its galaxy, the Milky Way, that takes 200 million years to complete.

29. Which of the following best describes the main subject of the passage?

(A) The various types of the Earth's motions
(B) Past changes in the Earth's position
(C) The predictability of the Earth's orbit
(D) Oddities of the Earth's rotation on its axis
(E) The Moon's gravitational effect on the Earth

30. The passage is most likely directed toward an audience of

(A) geologists
(B) astronauts
(C) aircraft navigators
(D) meteorologists interested in weather prediction
(E) persons with little technical knowledge of astronomy

31. Which of the following techniques does the author use in order to make the descriptions of motion clear?

I. Comparisons with familiar objects
II. Reference to geometric forms
III. Allusions to the works of other authors

(A) I only
(B) II only
(C) I and II only
(D) I and III only
(E) I, II, and III

32. The passage indicates that a single cycle of which of the following motions is completed in the shortest period of time?

(A) Nutation
(B) Precession
(C) The Earth's rotation on its axis
(D) The Earth's revolution around the Sun
(E) The Sun's movement around the hub of the Milky Way

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Poetry is not like reasoning, a power to be exerted according to the determination of the will. A person cannot say, "I will compose poetry." The greatest poet even cannot say it: for the mind in creation is as a fading coal, which some invisible influence, like an inconstant wind, awakens to transitory brightness: this power arises from within, like the color of a flower which fades and changes as it is developed, and the conscious portions of our natures are unprophetic either of its approach or its departure. Could this influence be durable in its original purity and force, it is impossible to predict the greatness of the results; but when composition begins, inspiration is already on the decline, and the most glorious poetry that has ever been communicated to the world is probably a feeble shadow of the original conception of the poet. I appeal to the great poets of the present day, whether it be not an error to assert that the finest passages of poetry are produced by labor and study. The toil and the delay recommended by critics, can be justly interpreted to mean no more than a careful observation of the inspired moments, and an artificial connection of the spaces between their suggestions by the intertexture of conventional expressions; a necessity only imposed by the limitedness of the poetical faculty itself.

Line
(5)
(10)
(15)
(20)

33. The comparison of "the mind in creation" to "a fading coal" touched by the wind (lines 4-5) emphasizes which of the following?

(A) The infinite variety of the intellect
(B) The inability of the mind to reason consistently
(C) The nourishment that poetry gives the mind
(D) The fleeting nature of inspiration
(E) The passionate quality of poetry

34. The author appeals to "the great poets of the present day" (line 16) to support the view that

(A) poets are made not born
(B) great poetry can be written only by people of great intellect
(C) familiarity with literary tradition is essential to poetic greatness
(D) modern poets are at least as talented as classical authors
(E) diligent effort and training do not produce the best poetry

35. By the "limitedness of the poetical faculty" (lines 23-24), the author means a poet's

(A) failure to comprehend the importance of study for poetic composition
(B) inability to sustain inspiration for long periods
(C) inability to understand the nature of inspiration
(D) reluctance to compose longer poetical works
(E) failure to make extensive use of traditional expression in poetry

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1

Even if history is viewed from other than equalitarian perspectives, and it is granted that differentials in economic rewards are morally justified and socially useful, it is impossible to justify the degree of inequality which complex societies inevitably create by the increased centralization of power that develops with more elaborate civilizations. The literature of all ages is filled with rational and moral justifications for these inequalities, but most of them are specious. If superior abilities and services to society deserve special rewards, it may be regarded as axiomatic that the rewards are always higher than the services warrant. No impartial authority determines the rewards. The people of power who control society grant these perquisites to themselves. Whenever special ability is not associated with power, as in the case of modern professionals, the excess of income over the average is ridiculously low in comparison with that of the economic overlords, who are the real centers of power in an industrial society.

Most rational and social justifications for unequal privilege are clearly afterthoughts. The facts are created by the disproportion of power that exists in a given social system. The justifications are usually dictated by the desire of the powerful to hide the nakedness of their greed and by the inclination of society to veil the brutal facts of human life from itself. This inclination is rather pathetic but understandable, since the facts of collective life easily rob the average individual of confidence in the human enterprise. The inevitable hypocrisy that is associated with all of the collective activities of the human race springs chiefly from this source: individuals have a moral code that makes the actions of society an outrage to their conscience. They therefore invent romantic and moral interpretations of the real facts, preferring to obscure rather than reveal the true character of their collective behavior. Sometimes they are as anxious to offer moral justifications for the brutalities from which they suffer as for those which they commit. The fact that the hypocrisy of group behavior expresses itself not only in terms of self-justification but in terms of moral justification of human behavior in general symbolizes one of the tragedies of the human spirit: its inability to conform its collective life to its individual ideals. As individuals, people believe that they ought to love and serve each other and establish justice toward each other. As economic and national groups, they take for themselves whatever their power can command.

36. The author refers to "modern professionals" (line 16) as examples of those who

(A) have the potential to become economic overlords
(B) are least likely to rebel against an unjust society
(C) can achieve incomes far in excess of the benefits they can give society
(D) lack the power to reward themselves excessively for their special abilities
(E) are unable to develop their special abilities for the greatest social good

37. According to the passage, those who would defend current levels of inequality in society are usually motivated by

(A) naïve idealism
(B) self-protective dishonesty
(C) an earnest concern for the well-being of others
(D) an obsessive desire for progress
(E) fear of reprisals from those in power

38. According to the passage, the origin of the "inevitable hypocrisy" (line 29) is

(A) a pervasive misunderstanding of the means by which inequalities in society can be reduced
(B) the dissimilarity in the moral codes of the individuals who constitute a large society
(C) the disparity between the individual's moral code and that of the larger society
(D) human inability to develop any sense of what a moral code should be
(E) the demands of an unselfish society on individuals who are innately selfish

39. The author's tone in the passage is that of a person attempting to

(A) urge the reader to take specific action
(B) ridicule previous analyses of an issue
(C) evaluate a theory without making a judgment
(D) explain an unfortunate and unrecognized truth
(E) defend a previously attacked theory

40. As described by the author, the average individual is all of the following EXCEPT

(A) defiant (B) compassionate
(C) hypocritical (D) self-deceptive
(E) unjustly treated

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

SECTION 3 Time—30 minutes 25 Questions

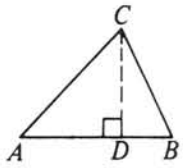
In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and fill in the corresponding oval on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definition of symbols:

$=$ is equal to	$\leqq$ is less than or equal to
$\neq$ is unequal to	$\geqq$ is greater than or equal to
$<$ is less than	$\parallel$ is parallel to
$>$ is greater than	$\perp$ is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

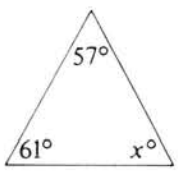
(1) area of $\triangle ABC = \frac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

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1. If $x + 3 = 13$, what does $x - 5$ equal?

   (A) 5 (B) 8 (C) 10 (D) 11 (E) 15



2. In the triangle above, $x =$

   (A) 59 (B) 60 (C) 61 (D) 62 (E) 63

3. If $y = 2x - 3$, then $-2y =$

   (A) $5x + 3$
   (B) $5x - 3$
   (C) $2x + 6$
   (D) $-4x - 6$
   (E) $-4x + 6$

4. If $x > -3$, then $x + 4$ could equal

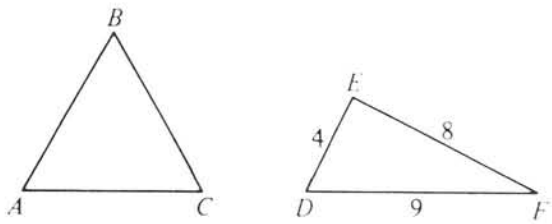
   (A) 2 (B) 1 (C) 0 (D) -6 (E) -8

5. The odometer of a new automobile functions improperly and registers only 2 miles for every 3 miles driven. If the odometer indicates 48 miles, how many miles has the automobile actually been driven?

   (A) 144 (B) 72 (C) 64 (D) 32 (E) 24

6. If $\sqrt{16} = 2^k$, what is the value of k?

   (A) 1 (B) 2 (C) 3 (D) 4 (E) 8



7. For the triangles above, the perimeter of $\triangle ABC$ equals the perimeter of $\triangle DEF$. If $\triangle ABC$ is equilateral, what is the length of side AB?

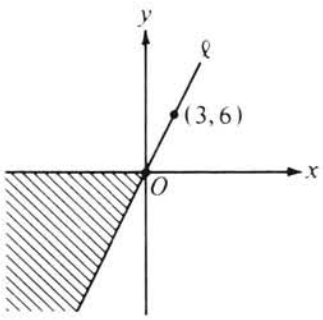
   (A) 4 (B) 5 (C) 7 (D) 9 (E) 15

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8. If $q < r < s < t$ where r and s are odd integers and q and t are even integers, what is the least possible value of $t - q$?

(A) 3 (B) 4 (C) 6 (D) 7 (E) 8



9. Of the following, which are the coordinates of a point that is located in the shaded region of the graph shown above?

(A) $(-4, -5)$ (B) $(-4, 5)$ (C) $(-5, -12)$
(D) $(-5, 10)$ (E) $(5, -10)$

---

10. If P was the number of Persian soldiers at the battle of Marathon and G was the number of Greek soldiers at that battle, which of the following equations can represent the statement "The number of Persian soldiers was 3 times the number of Greek soldiers"?

(A) $3P = 3G$
(B) $P = G + 3$
(C) $P + 3 = G$
(D) $G = 3P$
(E) $P = 3G$

Questions 11-13 refer to the following definition.

Let $\#$ be defined by the equation
$a \# b = ab + a + b$ for all numbers a and b.

11. $2 \# 5 =$

(A) 7 (B) 10 (C) 17 (D) 20 (E) 32

12. If $10 \# h = 98$, then $h =$

(A) 8
(B) 9
(C) 10
(D) 11
(E) 12

13. For what value of x is the statement $x \# y = x$ always true?

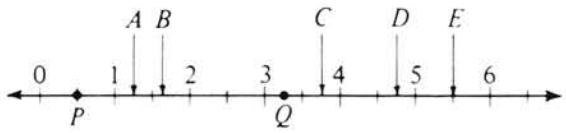
(A) -2
(B) -1
(C) 0
(D) 1
(E) 2

---

14. The sum of the digits of a two-digit integer is 12. If the units' digit is one-half the tens' digit, then the two-digit integer is

(A) 26
(B) 48
(C) 62
(D) 84
(E) 93

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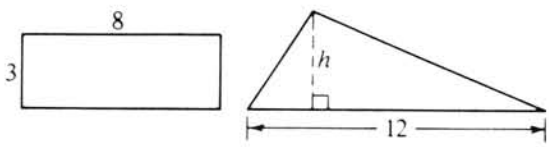
15. If P and Q are numbers on the number line above, which of the points shown best represents $P \times Q$?

(A) A (B) B (C) C (D) D (E) E

$$\frac{5.\blacksquare 81}{23} = N$$

16. In the fraction above, the four-digit numerator has one digit covered by ■. If N is a whole number, which of the following could be the value of N?

(A) 244 (B) 245 (C) 246
(D) 247 (E) 248



17. In the figures above, if the area of the rectangle is equal to the area of the triangle, then $h =$

(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

18. If the degree measures of the four angles of a quadrilateral are in the ratio 1 : 2 : 3 : 4, what is the degree measure of the largest angle?

(A) 36 (B) 90 (C) 144
(D) 160 (E) 180

19. A person paid a total of $700 for shirts priced at $9 and $13 each. If the person purchased two $13 shirts for every $9 shirt, what was the total number of shirts purchased?

(A) 20
(B) 23
(C) 40
(D) 60
(E) 68

20. If 1 jar + 5 cups = 1 tub and if 3 jars + 2 cups = 2 tubs, how many cups equal a tub?

(A) 8
(B) 11
(C) 13
(D) 15
(E) 18

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21. Points Q and R are on segment PS, which has length 24. If PQ is half as long as QS, and QR is 3 times as long as RS, how long is RS?

(A) 2
(B) 3
(C) 4
(D) 8
(E) 12

22. If United States imports increased 20 percent and exports decreased 10 percent during a certain year, the ratio of imports to exports at the end of the year was how many times the ratio at the beginning of the year?

(A) $\frac{12}{11}$

(B) $\frac{4}{3}$

(C) $\frac{11}{8}$

(D) $\frac{3}{2}$

(E) 2

23. If $xy = 1$, then y is the reciprocal of x. Which of the following equals the average (arithmetic mean) of x and its reciprocal?

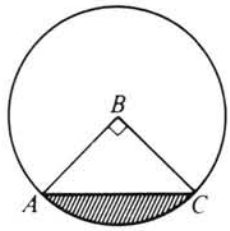
(A) $\frac{x^2 + 1}{2x}$

(B) $\frac{x + 1}{2x}$

(C) $\frac{x^2 + 2}{2x}$

(D) $\frac{2x^2 + 2}{x}$

(E) $\frac{x^2 + 1}{x}$



24. In the figure above, B is the center of the circle with radius 6. What is the area of the shaded region?

(A) 9π

(B) $36 - 9\pi$

(C) $36\pi - 18$

(D) $18 - \frac{9\pi}{2}$

(E) $9\pi - 18$

---

25. The sum of the first 1,000 positive integers is 500,500. What is the sum of the first 2,000 positive integers?

(A) 1,001,000
(B) 1,500,500
(C) 2,000,000
(D) 2,001,000
(E) 2,500,500

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SECTION 5 Time—30 minutes 35 Questions

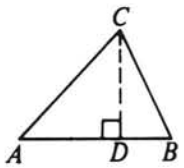
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Note: Figures that accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

1. Amelia has saved \$80 to buy a stereo that costs a total of \$220. If her take-home pay is \$4 an hour, how many hours will she have to work in order to have just enough to buy the stereo?

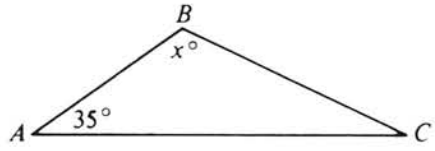
   (A) 35
   (B) 40
   (C) 45
   (D) 50
   (E) 55

2. On a certain map, the distance of 100 miles is represented by 1 inch. What is the distance represented by 2.4 inches on this map?

   (A) 200.4 mi
   (B) 204.0 mi
   (C) 225.0 mi
   (D) 233.3 mi
   (E) 240.0 mi

3. Three students have a total of 25 pencils among them. If the first student has 32 percent of the pencils and the second student has 48 percent of the pencils, how many pencils does the third student have?

   (A) 5 (B) 6 (C) 8 (D) 9 (E) 10



Note: Figure not drawn to scale.

4. In $\triangle ABC$ above, if $AB = BC$, then $x =$

   (A) 35 (B) 55 (C) 70 (D) 110 (E) 135

5. $\frac{5}{12}$ of $3 \times 6 \times 4 =$

   (A) 6 (B) 10 (C) 20 (D) 24 (E) 30

6. If $ax + 3 = 7$ and $bx + 5 = 12$, what is the value of $\frac{a}{b}$?

   (A) $\frac{3}{11}$ (B) $\frac{4}{7}$ (C) $\frac{7}{12}$ (D) $\frac{10}{17}$ (E) $\frac{3}{5}$

7. Twenty-eight identical boxes are stacked so that each layer above the bottom layer consists of 1 box fewer than the layer immediately below it. The top layer contains only 1 box. How many layers make up the stack?

   (A) Eight
   (B) Seven
   (C) Six
   (D) Five
   (E) Four

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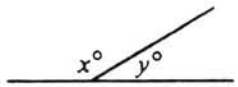
5

Questions 8-27 each consist of two quantities, one in Column A and one in Column B. You are to compare the two quantities and on the answer sheet fill in oval

A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

AN E RESPONSE WILL NOT BE SCORED.

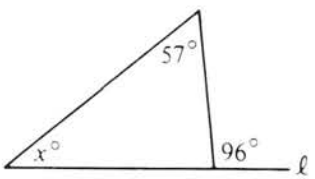
EXAMPLES

	Column A	Column B	Answers
E1.	2×6	$2 + 6$	● Ⓑ Ⓒ Ⓓ Ⓔ
E2.	 $180 - x$	y	Ⓐ Ⓑ ● Ⓓ Ⓔ
E3.	$p - q$	$q - p$	Ⓐ Ⓑ Ⓒ ● Ⓔ

Notes:

1. In certain questions, information concerning one or both of the quantities to be compared is centered above the two columns.
2. In a given question, a symbol that appears in both columns represents the same thing in Column A as it does in Column B.
3. Letters such as x, n, and k stand for real numbers.

	Column A	Column B
8.	$5^2 - 3^2$	2^2



9.	x	27

$x < 0$

10.	x	x^2
11.	Amount of interest earned in 2 months on x dollars deposited at 6% annual interest	\$10
12.	76×380	760×38



$y = 5$

$x : y = 3 : 1$

13.	$x + y$	20

	Column A	Column B

$(5 - r)s = 5s = 25$

14.	r	s

$x < y < z$

15.	y	$\frac{x + z}{2}$

a and b are positive integers.

$3a - 5b = 1$

16.	a	b
17.	The number of positive integers from 1 to 10^5 inclusive	The number of negative integers from -1 to $(-10)^5$ inclusive
18.	Area of a circle with diameter d	Area of a circle with radius $\frac{d}{2}$

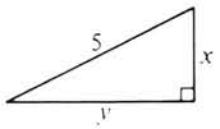
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SUMMARY DIRECTIONS FOR COMPARISON QUESTIONS

Answer: A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

AN E RESPONSE WILL NOT BE SCORED.

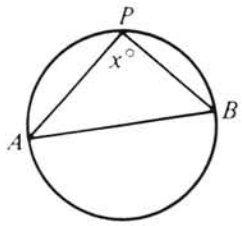
	Column A	Column B
	k is an integer greater than 1.	
19.	$k^2 + k^3$	k^4



	$y = 2x$	
20.	x	3

21.	The time it would take a car to travel 50 miles at an average speed of 55 miles per hour	The time it would take a car to travel 55 miles at an average speed of 50 miles per hour

	p, q, and r are consecutive integers and $pqr = 0$.	
22.	The greatest of p, q, and r	0



Note: Figure not drawn to scale.

	AB is any chord of the circle. P is a point on the circle.	
23.	x	90

	Column A	Column B
	$x \neq -1$	
24.	$\dfrac{4x^2 + 8x + 4}{(x + 1)^2}$	4

	A 2-kilogram solid cube has edges of length 10 centimeters. (Assume uniform density.)	
25.	The length of an edge of a 4-kilogram solid cube made from the identical material	20 centimeters

26.	$\dfrac{1}{1 - (0.6)^3}$	1

27.	The greatest power of 2 that is a factor of n, where $n = 2^2 \cdot 3 + 2^2 \cdot 7$	The greatest power of 2 that is a factor of k, where $k = 2 \cdot 3 + 2 \cdot 5$

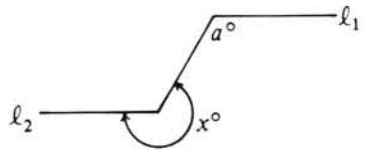
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5

Solve each of the remaining problems in this section using any available space for scratchwork. Then decide which is the best of the choices given and fill in the corresponding oval on the answer sheet.

28. If 1.39962 is rounded to the nearest thousandth, how many of its digits will remain unchanged?

(A) One (B) Two (C) Three
(D) Four (E) Five



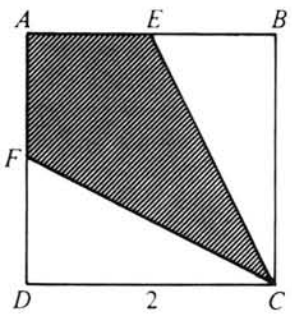
29. In the figure above, if $\ell_1 \parallel \ell_2$, what is the value of x in terms of a?

(A) $180 - a$
(B) $180 + a$
(C) $270 - a$
(D) $360 + a$
(E) $360 - a$

---

30. If 25 squares, each painted one of the solid colors red, green, yellow, or blue, are lined up side by side in a single row so that no two adjacent squares are the same color and there is at least one square of each color, what is the maximum possible number of blue squares?

(A) 9 (B) 10 (C) 11 (D) 12 (E) 13



31. In the figure above, $ABCD$ is a square with side of length 2. If E is the midpoint of line segment AB and F is the midpoint of line segment AD, then the area of quadrilateral $CFAE$ is

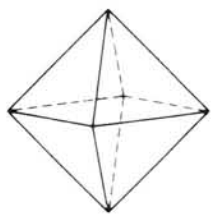
(A) 1 (B) $\frac{3}{2}$ (C) 2 (D) $\frac{7}{2}$ (E) 4

32. What is the result when the sum of the odd integers from 1 to 29, inclusive, is subtracted from the sum of the even integers from 2 to 30, inclusive?

(A) 0
(B) 1
(C) 15
(D) 16
(E) 30

33. If k and h are constants and $x^2 + kx + 7$ factors into $(x + 1)(x + h)$, what is the value of k?

(A) 0 (B) 1 (C) 7 (D) 8
(E) It cannot be determined from the information given.



34. The octahedron shown above is a solid figure whose eight equal faces are equilateral triangles, each with perimeter 5. The sum of the lengths of all the edges of the octahedron is

(A) $13\frac{1}{3}$ (B) 20 (C) $26\frac{2}{3}$
(D) 40 (E) 60

---

35. It takes Bill $2m$ minutes to complete form A, which has q questions, and $3m$ minutes to complete form B, which has $\frac{1}{2}q$ questions. If Bill answers all of the questions, his average (arithmetic mean) time to answer a form B question is how many minutes longer than his average time to answer a form A question?

(A) $\frac{4m}{q}$ (B) $\frac{10m}{3q}$ (C) $\frac{2m}{q}$
(D) $3mq$ (E) $\frac{1}{2}mq$

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

SECTION 6 Time—30 minutes 45 Questions

For each question in this section, choose the best answer and fill in the corresponding oval on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. RESTRAINT: (A) accuracy (B) secrecy (C) efficiency (D) freedom (E) protection
2. MUSTY: (A) fragrant (B) mobile (C) irregular (D) optional (E) tranquil
3. IRRITATE: (A) completely deny (B) apply fluid (C) strongly affect (D) work off (E) calm down
4. SUBTLE: (A) valuable (B) mellow (C) spacious (D) obvious (E) similar
5. INCENTIVE: (A) excuse (B) failure (C) discouragement (D) advice (E) interruption
6. FOCUSED: (A) distant (B) muddled (C) occasional (D) incomplete (E) proven
7. VIRTUOSO: (A) unprincipled schemer (B) unskilled beginner (C) idealistic reformer (D) staunch supporter (E) enthusiastic tutor
8. SERPENTINE: (A) serious (B) periodic (C) harsh (D) straight (E) tangential
9. CHASTEN: (A) communicate (B) reward (C) accompany (D) misjudge (E) disown
10. ORNATE: (A) nameless (B) austere (C) fragile (D) authentic (E) marred
11. ATROPHY: (A) justify (B) thrive (C) retreat (D) repel (E) insist
12. CULPABLE: (A) impossible (B) pleasing (C) insincere (D) reasonable (E) blameless
13. EFFACE: (A) entrap (B) condense (C) oppose (D) consider improper (E) make conspicuous
14. PENCHANT: (A) rigidity (B) disinclination (C) arrogance (D) levity (E) insensitivity
15. ECLECTIC: (A) of high price (B) of sound mind (C) of enormous merit (D) from concrete details (E) from one source

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that, when inserted in the sentence, best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

16. The job allowed no relaxation, but Jackson enjoyed ---- and welcomed the challenge of dealing with the variety of problems that came across her desk every day.

(A) leisure (B) monotony (C) pressure (D) privacy (E) inertia

17. Scientists are studying the birth and growth of thunderstorms to discover what causes the difference between showers that enable crops to ---- and ---- storms that cause floods and erosion.

(A) flourish. .violent (B) wither. .damaging (C) grow. .harmless (D) parch. .severe (E) multiply. .essential

18. The ballet stage is a bright, seemingly weightless world where gravity is continually being ---- by the dancers.

(A) prolonged (B) reapportioned (C) unbalanced (D) reflected (E) defied

19. A dictatorship ---- its citizens to be docile and finds it expedient to make outcasts of those who do not ----.

(A) forces. .rebel (B) expects. .disobey (C) requires. .conform (D) allows. .withdraw (E) forbids. .agree

20. Granted that McMurphy was frequently ----, it is still hard to believe in him as a profound political or economic thinker.

(A) arbitrary (B) taciturn (C) perfunctory (D) perceptive (E) intemperate

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6

Each passage below is followed by questions based on its content. Answer the questions following each passage on the basis of what is stated or implied in that passage.

Dr. Lloyd A. Hall (1894-1971) was a pioneer of American food chemistry, a field that burgeoned in the period between the two world wars. He was born in Elgin, Illinois, the grandson of a free Black man who settled in Chicago in 1837. Hall studied at Northwestern University and the University of Chicago, and was chief chemist and director of research at Griffith Laboratories from 1925 to his retirement in 1959.

Hall's interests in food technology and industrial chemistry were varied; he held over one hundred patents on products and methods he had invented. One area that occupied much of his research was the sterilization of foods and food additives. Even today, many people believe that spices help preserve foods. Hall found, to the contrary, that many of the spices marketed (such as cloves, cinnamon, ginger, paprika, allspice, and sage) were teeming with bacteria, molds, and yeasts. Even dried vegetables and herbs (onion powder and garlic powder) were guilty of contaminating food. Meat packers who added spices and seasonings to their meat products were actually contaminating them.

Hall undertook long and exhaustive research into the question of how to sterilize these foodstuffs effectively and at the same time preserve their appearance, quality, and flavor. Spices darkened when heated in air; they lost their flavor and aroma when subjected to evaporation or oxidation. Sterilizing spices with dry or moist heat above 240°F ruined the color and flavor of the spices so as to make them unmarketable.

After experimenting with a number of chemicals, Hall hit on ethylene oxide, a gas whose ability to kill insects was well known. Hall now used it to kill germs in foodstuffs. First, however, the chemical had to penetrate the foodstuff in order to reach the germs. This was difficult because the moisture and gases present on the surface and in the interior of the substances to be sterilized prevented the entrance of the ethylene oxide. Hall solved this problem by subjecting the foodstuff to a vacuum, a technique that removed moisture and rid the surfaces and interior of the foodstuff of any gases. In this way, the foodstuff was activated to take up the sterilizing gas. Hall then introduced the ethylene oxide into the vacuum chamber for the period of time necessary to sterilize the material; conditions of time and temperature differed according to the microorganism to be killed.

The introduction of sterilized spices to the meat-packing industry did much to revolutionize it. Hall's process was also applied to medicines, medical supplies, cosmetics, and other products. Sterilization with ethylene oxide became a very large business; Hall's method and its derivatives remain in general use today.

21. The passage is primarily concerned with

(A) describing a scientific advance achieved by Hall
(B) referring to scientific impasses that plagued Hall
(C) correcting a misconception about food preservation
(D) suggesting the breadth of Hall's career
(E) tracing the progress of food technology in this century

22. The passage suggests that, until Hall's breakthrough, all available methods of spice sterilization were impractical because they

(A) were too costly for use in mass production
(B) were suspected of posing health risks to consumers
(C) failed to yield results that were consistent
(D) made the product less attractive to the eye and palate
(E) sterilized only the surface of the spices

23. According to the passage, subjecting spices to a vacuum was crucial because this process made possible the

(A) prevention of further contamination by airborne matter
(B) extraction of the contaminants from the foodstuff
(C) precise control of conditions like time and temperature
(D) replication of the experiment by independent researchers
(E) impregnation of the foodstuff with ethylene oxide

24. It can be inferred that Hall's method for introducing ethylene oxide into foodstuffs worked because the gas

(A) occupied the spaces left vacant by the previously extracted gases and moisture
(B) was subjected to pressure many times that of the atmosphere
(C) displayed a natural affinity for any moisture-free substance
(D) became active when the temperature of the vacuum chamber was raised
(E) expanded in proportion to the time it remained in the vacuum chamber

25. The passage suggests that Hall's process was important for which of the following reasons?

I. It helped improve sanitary standards in the meat-processing industry.
II. It proved useful in areas unrelated to the food industry.
III. It enabled Hall to get funds for research into other problems.

(A) I only (B) I and II only (C) I and III only (D) II and III only (E) I, II, and III

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It is only recently, in the 1930's, that parents and teachers have ceased taking childhood and adolescence for granted. Rather than press the child into an inflexible educational mold, they have at last begun to fit education to the needs of the child. To this new task they were spurred by two forces: the growth of the science of psychology and the increasing, or at least more visible, maladjustments of youth. Psychologists alerted society to its need to gain knowledge about the ways in which children develop, about the stages through which they pass, and about the kinds of behavior adults might reasonably expect of a two-month-old baby or a two-year-old child. At the same time, the fulminations from the pulpit, the loudly voiced laments of the social philosopher, the records of juvenile courts and social agencies all demanded that something be done about the stage which science had named adolescence. The growing spectacle of a younger generation rejecting respected home standards and conventional religious values was a social phenomenon that worried even the least thoughtful among us. To adults, adolescents seemed cut adrift, without anchorage.

In American civilization, with its many immigrant strains, its dozens of conflicting standards of conduct, its hundreds of religious sects, and its shifting economic conditions, this unsettled, disturbed status of youth was more apparent than in the older, more settled civilizations. American conditions challenged the psychologist, the educator, and the social philosopher to propose acceptable explanations of the plight of children as they mature. Now a great mass of literature about adolescence is flooding bookshops as theoretical psychologists try to account for the restlessness of youth. Popular works by psychologists identify the various causes of the conflict and distress felt by young people, characterizing adolescence as the period in which idealism flowers and rebellion against authority waxes strong, a period when difficulties and conflicts are absolutely inevitable.

Many experimental psychologists, however, whose work is based on empirical data, do not subscribe to these views. Instead, they claim, "We have very little data. We are only just learning when a baby's eyes will first follow a light. How can we presume to define the needs of a developed personality, about which we know nothing?"

26. The author's primary purpose in this passage is to

(A) discuss a growth of interest in the problems of adolescence
(B) call for more psychological research in the area of adolescent behavior
(C) explore the ways of dealing with the problems of adolescents
(D) examine the various sources of conflict that confront adolescents
(E) define adolescence on the basis of psychological research

27. Which of the following does the author consider a major advance resulting from the increasing awareness of the problems of adolescence?

(A) Increased sensitivity in juvenile court systems
(B) A decline in the number of psychologically disturbed youths
(C) More flexible standards in religious organizations
(D) The revision of educational practices
(E) The growing influence of experimental psychologists

28. The phrase "cut adrift, without anchorage" (lines 21-22) is used in the passage to mean that adolescents

(A) receive insufficient advice from parents
(B) thrive on rebellion and risk
(C) have difficulty earning respect from adults
(D) espouse philosophies that they will later regret
(E) have no standards to replace the ones they rejected

29. It can be inferred from the passage that experimental psychologists are more interested than theoretical psychologists in

(A) carefully examining all aspects of human behavior before attempting to draw conclusions about adolescence
(B) formulating theories about the role of the adolescent in contemporary society
(C) sincerely helping adolescents learn to identify and cope with their most profound problems
(D) administering help to individuals rather than advising groups of people
(E) observing adolescent behavior from a historical point of view

30. The author's attitude toward the problems of adolescence can best be described as one of

(A) increasing apprehension
(B) objective reflection
(C) deep-rooted frustration
(D) condescending amusement
(E) open confusion

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6

Select the word or set of words that best completes each of the following sentences.

31. Are we to become mere ----, watching the panorama without personal involvement, our senses dulled, our capacity to ---- undeveloped?

(A) captives. .submit (B) imitators. .observe
(C) bystanders. .act (D) entertainers. .relax
(E) disciples. .assist

32. Although laboratory scientists often find themselves performing routine functions, their training should indicate to them ways in which their knowledge can be applied in ---- manner.

(A) an overt (B) a practical (C) an original
(D) a professional (E) a systematic

33. Because the situation called for immediate action, I ---- my natural inclination to consider the issues with ---- before making a decision.

(A) controlled. .improvidence
(B) followed. .circumspection
(C) repudiated. .expediency
(D) resisted. .deliberation
(E) accepted. .imprudence

34. In the nineteenth century many literary critics saw themselves as stern, authoritarian figures defending society against the ---- of those ---- beings called authors.

(A) depravities. .wayward
(B) atrocities. .exemplary
(C) merits. .ineffectual
(D) kudos. .antagonistic
(E) indictments. .reticent

35. Because he felt intimidated in his new position, he was ---- divulging his frank opinions of company proposals.

(A) scurrilous about (B) candid in
(C) chary of (D) fervid about
(E) precipitate in

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

36. BLUEPRINT : BUILDING :: (A) trunk : tree
(B) skin : hand (C) watch : time
(D) photograph : camera (E) pattern : dress

37. DECISION : JUDGE :: (A) evidence : jury
(B) verdict : defendant (C) examination : teacher
(D) diagnosis : doctor (E) election : candidate

38. ALUMINUM : METAL :: (A) alloy : element
(B) steel : tin (C) hydrogen : water
(D) oxygen : gas (E) diamond : ring

39. REHEARSAL : PLAY :: (A) draft : essay
(B) manual : process (C) applause : performance
(D) recital : concert (E) journal : news

40. TALON : HAWK :: (A) horn : bull
(B) fang : snake (C) claw : tiger
(D) tail : monkey (E) shell : tortoise

41. APOLOGY : RUEFUL ::
(A) confession : inquisitive
(B) request : grateful
(C) recommendation : censorious
(D) boast : proud
(E) taunt : timid

42. DEFECTOR : CAUSE :: (A) counterfeiter : money
(B) deserter : army (C) critic : book
(D) advertiser : sale (E) intruder : meeting

43. STUPEFACTION : ALERT ::
(A) vexation : curious
(B) contention : grave
(C) perplexity : courteous
(D) apprehension : optimistic
(E) enlightenment : wise

44. CUPIDITY : WEALTH :: (A) affection : passion
(B) intelligence : wit (C) mourning : death
(D) gluttony : food (E) audacity : fear

45. TACIT : WORDS ::
(A) visible : scenes (B) inevitable : facts
(C) colorful : hues (D) suspicious : clues
(E) unanimous : disagreements

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

Correct Answers for Scholastic Aptitude Test Form Code 5E

VERBAL		MATHEMATICAL	
Section 1	**Section 6**	**Section 3**	**Section 5**
1. A	1. D	1. A	1. A
2. C	2. A	2. D	2. E
3. E	3. E	3. E	3. A
4. D	4. D	4. A	4. D
5. B	5. C	5. B	5. E
6. E	6. B	6. B	6. B
7. B	7. B	7. C	7. B
8. C	8. D	8. B	*8. A
9. E	9. B	9. A	*9. A
10. E	10. B	10. E	*10. B
11. D	11. B	11. C	*11. D
12. B	12. E	12. A	*12. C
13. A	13. E	13. B	*13. C
14. B	14. B	14. D	*14. B
15. A	15. E	15. B	*15. D
16. D	16. C	16. D	*16. A
17. A	17. A	17. D	*17. C
18. B	18. E	18. C	*18. C
19. C	19. C	19. D	*19. B
20. E	20. D	20. C	*20. B
21. B	21. A	21. C	*21. B
22. A	22. D	22. B	*22. D
23. E	23. E	23. A	*23. D
24. B	24. A	24. E	*24. C
25. C	25. B	25. D	*25. B
26. C	26. A		*26. A
27. B	27. D		*27. B
28. C	28. E		28. A
29. A	29. A		29. E
30. E	30. B		30. E
31. C	31. C		31. C
32. C	32. C		32. C
33. D	33. D		33. D
34. E	34. A		34. B
35. B	35. C		35. A
36. D	36. E		
37. B	37. D		
38. C	38. D		
39. D	39. A		
40. A	40. C		
	41. D		
	42. B		
	43. D		
	44. D		
	45. E		

*Indicates four-choice questions. (All of the other questions are five-choice.)

The Scoring Process

Machine-scoring is done in three steps:

- *Scanning.* Your answer sheet is "read" by a scanning machine and the oval you filled in for each question is recorded on a computer tape.
- *Scoring.* The computer compares the oval filled in for each question with the correct response. Each correct answer receives one point; omitted questions do not count toward your score. For each wrong answer, a fraction of a point is subtracted to correct for random guessing. For questions with five answer choices, one-fourth of a point is subtracted for each wrong response; for questions with four answer choices, one-third of a point is subtracted for each wrong response. The SAT-verbal test has 85 questions with five answer choices each. If, for example, a student has 44 right, 32 wrong, and 9 omitted, the resulting raw score is determined as follows:

$$44 \text{ right} - \frac{32 \text{ wrong}}{4} = 44 - 8 = 36 \text{ raw score points}$$

  Obtaining raw scores frequently involves the rounding of fractional numbers to the nearest whole number. For example, a raw score of 36.25 is rounded to 36, the nearest whole number. A raw score of 36.50 is rounded upward to 37.
- *Converting to reported scaled score.* Raw test scores are then placed on the College Board scale of 200 to 800 through a process that adjusts scores to account for minor differences in difficulty among different editions of the test. This process, known as equating, is performed so that a student's reported score is not affected by the edition of the test taken nor by the abilities of the group with whom the student takes the test. As a result of placing SAT scores on the College Board scale, scores earned by students at different times can be compared. For example, an SAT-verbal score of 400 on a test taken at one administration indicates the same level of developed verbal ability as a 400 score obtained on a different edition of the test taken at another time.

How to Score the Test

SAT-Verbal Sections 1 and 6

Step A: Count the number of correct answers for *section 1* and record the number in the space provided on the worksheet on the next page. Then do the same for the incorrect answers. (Do not count omitted answers.) To determine subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for *section 6* and record the numbers in the spaces provided on the worksheet. To determine subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: To obtain C, add subtotal A to subtotal B, keeping any decimals. Enter the resulting figure on the worksheet.

Step D: To obtain D, your raw verbal score, round C to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step E: To find your reported SAT-verbal score, look up the total raw verbal score you obtained in step D in the conversion table on page 142. Enter this figure on the worksheet.

SAT-Mathematical Sections 3 and 5

Step A: Count the number of correct answers and the number of incorrect answers for *section 3* and record the numbers in the spaces provided on the worksheet. To determine the subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for the *five-choice questions (questions 1 through 7 and 28 through 35) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: Count the number of correct answers and the number of incorrect answers for the *four-choice questions (questions 8 through 27) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal C, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{3} = \text{subtotal C}$$

Step D: To obtain D, add subtotal A, subtotal B, and subtotal C, keeping any decimals. Enter the resulting figure on the worksheet.

Step E: To obtain E, your raw mathematical score, round D to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step F: To find your reported SAT-mathematical score, look up the total raw mathematical score you obtained in E in the conversion table on page 142. Enter this figure on the worksheet.

SAT SCORING WORKSHEET

SAT-Verbal Sections

A. Section 1: _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal A

B. Section 6: _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal B

C. Total unrounded raw score (Total A + B) _____ C

D. Total rounded raw score (Rounded to nearest whole number) _____ D

E. SAT-verbal reported scaled score (See the conversion table on page 142.) [ ] SAT-verbal score

SAT-Mathematical Sections

A. Section 3: _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal A

B. Section 5: Questions 1 through 7 and 28 through 35 (5-choice) _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal B

C. Section 5: Questions 8 through 27 (4-choice) _____ (no. correct) − ⅓ (_____) (no. incorrect) = _____ subtotal C

D. Total unrounded raw score (Total A + B + C) _____ D

E. Total rounded raw score (Rounded to nearest whole number) _____ E

F. SAT-mathematical reported scaled score (See the conversion table on page 142.) [ ] SAT-math score

Score Conversion Table
Scholastic Aptitude Test
Form Code 5E

Raw Score	College Board Reported Score		Raw Score	College Board Reported Score	
	SAT-Verbal	SAT-Math		SAT-Verbal	SAT-Math
85	800		40	460	600
84	780		39	450	590
83	760		38	450	580
82	750		37	440	570
81	740		36	430	570
80	730		35	430	560
79	720		34	420	550
78	710		33	420	540
77	700		32	410	530
76	700		31	400	520
75	690		30	400	510
74	680		29	390	500
73	670		28	380	490
72	670		27	370	480
71	660		26	370	470
70	660		25	360	460
69	650		24	350	450
68	640		23	350	440
67	640		22	340	440
66	630		21	330	430
65	620		20	330	420
64	620		19	320	410
63	610		18	310	400
62	600		17	310	390
61	600		16	300	380
60	590	800	15	290	370
59	580	780	14	290	370
58	580	760	13	280	360
57	570	750	12	270	350
56	560	740	11	260	340
55	560	730	10	260	330
54	550	720	9	250	330
53	540	710	8	240	320
52	540	700	7	240	310
51	530	690	6	230	300
50	530	680	5	220	290
49	520	670	4	220	280
48	510	670	3	210	270
47	510	660	2	200	270
46	500	650	1	200	260
45	490	640	0	200	250
44	490	630	−1	200	240
43	480	620	−2	200	230
42	470	620	−3	200	220
41	470	610	−4	200	210
			−5 or below	200	200